RESEARCH METHODS

TITLE PROPOSAL

RESEARCHER:	SARMIENTO, DAVID DATU N.
Section:	3A
Proposed Research Title:	FlexiQueue: A Universal Queue Management System for Multi-Service Government Operations
Description: *(minimum of 200 words)*	FlexiQueue is a configurable queue management system designed to address the operational challenges faced by the Municipal Social Welfare and Development Office (MSWDO) and similar government agencies that operate multiple assistance programs with varying service requirements and process flows. Currently, MSWDO offices struggle to manage different programs such as cash assistance distributions, Pantawid Pamilya beneficiary verification, disaster relief operations, and social pension releases, each requiring different documents, verification steps, and service flows. Existing queue systems are either too rigid, forcing all programs into identical workflows, or entirely manual, relying on physical number tickets and verbal coordination that break down under high volume and create opportunities for confusion and favoritism. FlexiQueue provides MSWDO with a single platform that can be configured according to specific program requirements without requiring code changes or external technical support. The system uses a table-based architecture that mirrors actual government service operations, where services are organized around physical service stations rather than abstract queue lines. This design allows administrators to create and modify programs rapidly, enabling adaptation to new policies, emergency response situations, or seasonal program variations. The system is designed with portability in mind, capable of being deployed on compact hardware such as single-board computers paired with portable wireless access points, making it suitable for both permanent office installations and temporary distribution sites in remote barangays or emergency relief operations where full infrastructure may not be available. The system uses a stub-driven approach where clients receive privacy-protecting queue stubs with randomly generated aliases like "A1" or "B1" upon arrival. These stubs are pre-printed and exist independently of any specific program, allowing reuse across different MSWDO initiatives. When a client arrives, a routing clerk scans the stub and registers it to the active program, automatically creating the required process checklist based on program configuration. The stub then tracks

	the client's progress through multiple service tables and process types until all requirements are completed. When launching a new program, the MSWDO administrator configures which processes are required (such as consent forms, personal data sheets, interviews, ID verification, or cash claim forms), sets up service tables with appropriate staff assignments and capacity limits, and establishes flow rules that guide clients through required steps while allowing staff to override or adapt the flow based on real-time conditions. The system supports complex scenarios where a single program deployment can have Table 1 with one staff member assisting two clients simultaneously on forms, Table 2 with two staff members conducting interviews, and Table 3 with three staff members processing claims for six clients at once. Some clients may only need verification and release, while others require additional steps, and the system adapts without requiring modifications. Three device types support operations: Queue Devices at each table display currently served stubs with audio-visual calling, implementable as staff smartphones or dedicated displays; Scanner Devices allow staff to scan stub barcodes using smartphone cameras to view status, assign clients to tables, and mark processes complete; Queue Informant Devices function as self-service kiosks where clients independently scan stubs to view their current table assignment, pending processes, completion status, and estimated position without staff assistance. Flow configuration balances structured guidance with operational flexibility. Administrators define flow rules specifying process sequences and dependencies, but staff can override configured flows when necessary, with all overrides logged for accountability. The system operates fully offline using local network connectivity, ensuring reliability in remote locations without internet. All client movements, process completions, staff actions, and flow overrides are logged comprehensively for COA compliance and operational analysis. When program end dates are reached, all stubs registered to that program automatically expire, preventing reuse and maintaining data integrity across program cycles. By providing MSWDO with a single, portable platform that adapts to diverse program requirements, FlexiQueue enables more responsive, transparent, and efficient service delivery to Filipino beneficiaries who depend on government assistance, whether in the main office or remote distribution sites.

Objectives:	• To develop a table-based queue management platform that allows MSWDO administrators to configure service delivery for multiple assistance programs without requiring system code changes or external technical support. • To design a portable, offline-first system architecture capable of deployment on compact hardware suitable for both permanent office installations and temporary distribution sites in remote locations. • To design a process configuration system that allows administrators to define required forms, verification steps, and service flows specific to each MSWDO program while maintaining a reusable master catalog of process types. • To implement a stub-driven tracking mechanism that protects client privacy through random alias generation while maintaining full audit traceability of all service transactions and client movements across program lifecycle. • To create a flow engine that balances structured process guidance with staff override capabilities, allowing adherence to standard procedures while accommodating exceptions and special cases that commonly occur in government service delivery. • To establish device types for different operational roles including table queue devices for client calling, scanner devices for routing and process tracking, and informant devices for self-service status checking, with support for both simple smartphone implementations and optional IoT display enhancements. • To develop a comprehensive audit trail system that logs all client-service interactions, staff actions, process completions, and flow overrides for accountability, COA compliance verification, and operational analysis.
SDG: *(Mention the SDG and explain the alignment to the proposed title, minimum of 300 words)*	FlexiQueue primarily aligns with Sustainable Development Goal 16: Peace, Justice, and Strong Institutions, particularly Target 16.6 which emphasizes developing effective, accountable, and transparent institutions at all levels, and Target 16.7 which focuses on ensuring responsive, inclusive, participatory, and representative decision-making at all levels. FlexiQueue directly supports these targets by addressing systemic inefficiencies in MSWDO service delivery that disproportionately affect vulnerable populations who depend on

	government assistance programs for survival and livelihood support. In the Philippine context, MSWDO serves millions of beneficiaries annually through programs like Pantawid Pamilyang Pilipino Program (4Ps), Assistance to Individuals in Crisis Situation (AICS), Social Pension for Indigent Senior Citizens, and disaster response operations. Long wait times, unclear queuing processes, lack of real-time status visibility, and inconsistent service flows create significant barriers, particularly for elderly beneficiaries, persons with disabilities, pregnant women, and those with limited literacy who struggle to navigate complex government procedures. The portability of FlexiQueue strengthens institutional effectiveness by enabling MSWDO to bring services directly to remote barangays or emergency distribution sites, ensuring that even the most geographically isolated beneficiaries receive organized, transparent service delivery. FlexiQueue's privacy-protecting stub system with random aliases ensures that beneficiaries are served based on objective arrival time and process requirements rather than personal recognition or staff favoritism, directly supporting institutional accountability. The queue informant devices empower beneficiaries to independently verify their status and remaining requirements without staff assistance, reducing anxiety and promoting dignity in the service experience. This transparency mechanism supports Target 16.6 by making service delivery visible and verifiable. The system's configurability supports institutional effectiveness by allowing MSWDO offices to adapt service delivery to actual operational conditions and program requirements. When MSWDO launches new assistance programs or modifies eligibility requirements, offices can independently configure processes, define flow rules, and assign personnel without requiring system modification or external technical support. This operational autonomy strengthens institutional capacity and responsiveness to changing beneficiary needs and policy directives. FlexiQueue's comprehensive audit logging creates transparent records of every client movement, process completion, staff action, and flow override decision. These immutable logs can be reviewed by supervisors, Commission on Audit (COA) auditors, or oversight bodies investigating complaints, helping prevent corruption opportunities, supporting evidence-based management decisions, and building public trust in government service delivery.

	Secondary alignment exists with SDG 10: Reduced Inequalities, as the system specifically accommodates varying levels of beneficiary capabilities. Queue informant devices provide accessible status checking that reduces reliance on verbal communication, benefiting those with hearing difficulties or language barriers. The random alias system eliminates potential discrimination based on recognizable names that might indicate ethnicity, religious affiliation, or social status, ensuring equitable treatment regardless of background. The system also supports SDG 9: Industry, Innovation, and Infrastructure by demonstrating how appropriate technology can strengthen public service infrastructure in resource-constrained environments. The offline-first architecture with scalable display integration, combined with portable deployment capability on compact hardware, ensures that even offices in remote barangays with unreliable electricity or internet connectivity can implement modern queue management. The use of readily available devices like smartphones reduces hardware costs and technical barriers to adoption. Offices can start with simple smartphone-based deployments and progressively enhance their setup as resources become available, demonstrating a practical path for digital transformation in Philippine government social welfare operations. Overall, FlexiQueue strengthens MSWDO's institutional effectiveness, promotes equitable service access regardless of location, ensures operational transparency, and empowers both government staff and beneficiaries through technology design that respects the realities of Philippine government operations and the dignity of vulnerable populations seeking assistance.
Client/Recipient Agency/Office:	**Municipal Social Welfare and Development Office (MSWDO) - LGU Tagudin**
Contact Person:	**Contact Person:** Laarni L. Acosta
Rank:	Department Head

Prepared by:

DAVID DATU N. SARMIENTO
Researcher

Approved:

JOY G. BEA, DIT
Research Method Instructor